

Sixing Chen

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EDUCATION

New York University

Ph.D. in Cognition & Perception

New York, NY

Sep. 2023 – Present

Peking University

B.S. in Cognitive Science

Beijing, China

Sep. 2018 – May. 2023

EXPERIENCE

AI Safety Research Fellow, Cambridge Boston Alignment Initiative

Jun. 2026 – Present

Mentor: Natalie Shapira

- Identified circuit-level causal mechanisms underlying sycophancy in LLMs.
- Developed an inference-time causal intervention method that eliminates sycophancy in LLMs.

Doctoral Researcher, New York University

Sep. 2023 – Present

Advisor: Marcelo Mattar

- Investigated whether and how chain-of-thought reasoning implements planning in LLMs during large-scale strategic gameplay.
- Built deep reinforcement learning agents to model and predict eye-tracking data and single-neuron recordings during reasoning and decision-making.
- Developed a mathematical theory explaining how efficient planning can be implemented by mental simulation.

Undergraduate Research Assistant, Georgia Institute of Technology

Jan. 2022 – May 2023

Advisor: Dobromir Rahnev

- Designed behavioral experiments to study how cognitive factors shape perceptual decision-making.
- Analyzed large-scale behavioral datasets to characterize how humans report confidence in their decisions.

Undergraduate Research Assistant, Peking University

Sep. 2020 – May 2023

Advisor: Hang Zhang

- Developed computational models of how humans represent temporal uncertainty and how those representations shape motor preparation.

PUBLICATIONS & PREPRINTS

Chen, S., Li, J. A., Cakir, S., Akcali, S., Lee, K., & Mattar, M. G. (2026). Extracting Search Trees from LLM Reasoning Traces Reveals Myopic Planning. *Advances in Neural Information Processing Systems (NeurIPS)*.

Chen, S.*, Callaway, F.*, Kumar, S., Lupkin, S. M., Wallis, J. D., McGinty, V. B., Rich, E. L., & Mattar, M. G. (2026). Learning to select computations in recurrent neural circuits. *bioRxiv*.

Chen, S., Bocheva, M., & Rahnev, D. (2026). Signatures proposed to index perceptual effects emerge in a purely cognitive task. *Psychonomic Bulletin & Review*, 33(3), 75.

Nicholas, J., **Chen, S.**, & Mattar, M. G. (2026). Flexible decisions arise from resource-rational memory sampling. *bioRxiv*.

Chen, S.*, Jensen, K. T.*, & Mattar, M. G. (2025). Rational decisions in multi-step environments with few rollouts. *PsyArXiv*.

Gao, Y., **Chen, S.**, & Rahnev, D. (2025). Dynamics of sensory and decisional biases in perceptual decision making: Insights from the face distortion illusion. *Psychonomic Bulletin & Review*, 32(1), 317–325.

Chen, S., Jensen, K. T., & Mattar, M. G. (2024). Some and done? Temporally extended decisions with very few rollouts. *Proceedings of the Annual Meeting of the Cognitive Science Society*, Vol. 46.

Chen, S., & Rahnev, D. (2023). Confidence response times: Challenging postdecisional models of confidence. *Journal of Vision*, 23(7), 11–11.

CONFERENCE PRESENTATIONS

Chen, S., Callaway, F., Kumar, S., Wallis, J. D., Rich, E. L., & Mattar, M. G. (Aug. 2026). Learning to select computations in recurrent circuits. *Cognitive Computational Neuroscience (CCN)*, New York, NY, United States.

Chen, S., Callaway, F., Kumar, S., Wallis, J. D., Rich, E. L., & Mattar, M. G. (Apr. 2026). Learning to select computations in recurrent circuits. *Manhattan Area Learning Meeting (MALM)*, New York, NY, United States. [Oral Presentation]

Chen, S., Callaway, F., Kumar, S., Wallis, J. D., Rich, E. L., & Mattar, M. G. (Mar. 2026). Learning to select computations in recurrent circuits. *Computational and Systems Neuroscience (COSYNE)*, Lisbon, Portugal.

Chen, S., Callaway, F., Kumar, S., & Mattar, M. G. (Jun. 2025). Meta-learning of human-like planning strategies. *Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, Dublin, Ireland. [Oral Presentation]

Chen, S., Jensen, K. T., & Mattar, M. G. (Jul. 2024). Some and done? Temporally extended decisions with very few rollouts. *Annual Meeting of the Cognitive Science Society (CogSci)*, Rotterdam, Netherlands.

Chen, S., Jensen, K. T., & Mattar, M. G. (Mar. 2024). Some and done? Temporally extended decisions with very few rollouts. *Computational and Systems Neuroscience (COSYNE)*, Lisbon, Portugal.

Chen, S., Wang, M., & Zhang, H. (Nov. 2022). The alternation of human temporal beliefs between different possibilities. *Society for Neuroscience (SfN)*, San Diego, CA, United States.

TEACHING EXPERIENCE

New York University

Teaching Assistant, Computational Programming in Psychological Sciences

Feb. 2026 – May. 2026

Teaching Assistant, Lab for Cognition and Perception

Feb. 2025 – May. 2025

Peking University

Teaching Assistant, Computational Modeling for Psychology and Neuroscience

Feb. 2022 – Jun. 2022

AWARDS & SCHOLARSHIPS

AI Safety Research Fellowship, Cambridge Boston Alignment Initiative

2026

Engberg Fellowship, New York University

2023

Fenjiu Group Public Scholarship, Peking University

2021

TECHNICAL SKILLS

Programming: Python, JavaScript, R, MATLAB

ML / DL: PyTorch, TensorFlow, scikit-learn

LLM: vLLM, HuggingFace Transformers, NNSight

Tools: Git, Linux, SLURM, Jupyter